Improved Upper Bounds on the Size of Condorcet Domains

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Local Combinatorica draft, 29 October 2026

Abstract

We reduce full Condorcet domains to adjacent-swap components defined by extreme never-conditions and fixed pair comparisons. Their deletion sums add to the original deletion sum plus its insertion gaps. Combining this identity with Moros's gap inequality and an exact finite-window estimate gives $f(n) \le 32n\, f(n-1)/(11n+8)$, hence $f(n) = O(n^{-8/11}(32/11)^n)$. The same argument recovers the peak-pit growth bound $8/3$. Ordered concatenation removes the window boundary loss. The finite certificate and exact checking programs are supplied separately.

1 Introduction

A *Condorcet domain* is a family of linear orders for which pairwise majority is transitive for every odd voter profile. Let $f(n)$ be its maximum possible size on n alternatives. By Sen's characterization [2], each triple must have an alternative that never occupies one of its three positions. We write $x\mathrm{N}p$ for such a *never-condition*. A domain is *peak-pit* if every triple has a never-first or never-last condition; let $h(n)$ be the corresponding maximum. No common axis is required.

Raz [3] proved that $f(n)$ is at most exponential. Moros's manuscript [4] introduced the deletion-and-gap framework used here and obtained explicit general bases $56/19$ and $1336/455$. It also proved the peak-pit base $8/3$, using finite windows and potential cancellation. Our numerical improvement concerns the unrestricted bound; the peak-pit base is recovered within the same structural argument. For background and constructions, see [5, 6, 7].

Theorem 1.1. *For $n \ge 3$,*

$$f(n) \le \frac{32n}{11n+8} f(n-1), \qquad f(n) \le 4(32/11)^{n-3}, \tag{1}$$

$$h(n) \le \frac{8}{3} h(n-1), \qquad h(n) \le 4(8/3)^{n-3}. \tag{2}$$

The limits of $f(n)^{1/n}$ and $h(n)^{1/n}$ exist and are at most $32/11 = 2.9090909\ldots$ and $8/3$, respectively. Moreover, $f(n) = O(n^{-8/11}(32/11)^n)$.

The structural step is to reduce general domains to families defined by extreme never-conditions and fixed pair comparisons. Write D_{-x} for the distinct orders left after deleting x, and put

$$M = |D|, \qquad \Sigma(D) = \sum_x |D_{-x}|.$$

Every deletion preserves the Condorcet property, so a maximum domain satisfies $\Sigma(D) \le n f(n-1)$. For a full domain D, we prove that its adjacent-swap components K belong to the extreme-law class and satisfy

$$\sum_K \Sigma(K) = \Sigma(D) + G(D),$$

where $G(D)$ counts gaps between runs of insertion positions. This component reduction is the main structural contribution of the paper. It builds on Moros's framework and uses the median structure studied by Puppe and Slinko [8].

The extreme-law estimate $\Sigma(K) \geq 3n|K|/8$ therefore gives $\Sigma(D) + G(D) \geq 3n|D|/8$. Moros's maximal-domain gap inequality $G(D) \leq 3\Sigma(D) - (n+1)|D|$ then yields

$$\frac{3}{8}n|D| \leq \Sigma(D) + G(D) \leq 4\Sigma(D) - (n+1)|D|, \qquad \Sigma(D) \geq \frac{11n+8}{32}|D|. \tag{3}$$

Proposition 2.3 records the same transfer with an arbitrary uniform density input. Sections 2 and 3 give the structural and gap arguments. Section 4 retains the finite lemma, its exceptional case and coverage proof; Section 5 removes the boundary loss. Exact certificate tables, witness files and reproduction instructions are separate supplementary material.

2 Fibres and adjacent-swap components

A *full domain* is the set $D(N)$ of all orders satisfying one specified never-condition on each triple. Every inclusion-maximal Condorcet domain is full: choose a common condition on each triple, and use maximality. In auxiliary arguments we also allow fixed pair comparisons. Let $\mathcal{F}_n$ be the nonempty full domains defined by one extreme condition (xN1 or xN3) per triple and any fixed pairs. Pair conditions are necessary because conditioning an outside order can create them inside a window.

Fix x and $P \in D_{-x}$. Its *insertion fibre* is

$$F(x, P) = \{t \in \{0, \ldots, n-1\} : P[x:t] \in D\},$$

where $P[x:t]$ inserts x with t alternatives above it. If this set has r nonempty consecutive runs, it contributes $r - 1$ to $G(D)$. Let $A(D)$ count enabled adjacent swaps over every order and position; thus each undirected edge is counted twice.

Lemma 2.1. *For every finite family of orders,*

$$\Sigma(D) + G(D) = n|D| - A(D). \tag{4}$$

Every fibre of a domain in $\mathcal{F}_n$ is an interval, so $G(D) = 0$ there.

Proof. A fibre with m occupied slots and r runs has $m - r$ neighbouring occupied pairs. Each corresponds to moving its distinguished alternative down by one slot. Summed over all distinguished alternatives, these moves count every directed swap exactly once. Hence $A = nM - \sum_F r$, and $\sum_F r = \Sigma + G$. For an extreme law, the admissible insertion slots form an interval; the same is true of a fixed comparison. Intersect these intervals. □

Theorem 2.2 (Component reduction)**.** *Every adjacent-swap component K of a full Condorcet domain belongs to $\mathcal{F}_n$, and*

$$\sum_K \Sigma(K) = \Sigma(D) + G(D). \tag{5}$$

Proof. A full domain is closed under coordinatewise majority of pair comparisons. This is the median structure of closed Condorcet domains studied by Puppe and Slinko [8]; we give the component argument here. Indeed, the Condorcet property makes that majority transitive, and every never-condition is preserved by majority. For never-first and never-last this follows from the intersection

of two subsets of size two among three voters; never-middle says that the distinguished alternative's two comparisons agree.

Let U be all comparisons constant on K. We first show that K contains every order of D agreeing with U. Otherwise choose such a τ outside K and $\sigma \in K$ closest to it in comparison distance. A differing comparison is not constant on K, so some $\rho \in K$ agrees with τ there. Project a path from σ to ρ by $v \mapsto \mathrm{maj}(\sigma, \tau, v)$. The images remain in D, and consecutive images differ in at most one comparison. If two linear orders differ in only one pair comparison, that pair is adjacent in both orders. Thus the image path stays in K, while the final image is closer to τ, a contradiction.

A selected never-middle law xN2 forces the comparisons of x with the other two alternatives to agree. Neither can change along an adjacent-swap path, so both are fixed on K. Replace this law by the extreme law implied by their common value, keep the other extreme laws, and impose U. The resulting full solution set is exactly K. Thus $K \in \mathcal{F}_n$.

An original fibre meets each component in either no slots or one run: it is an interval within a component, and consecutive slots are joined by swaps. Its r runs are consequently counted r times in the component deletion sums. Summing gives (5). □

Proposition 2.3 (Transfer of a density estimate)**.** *Fix $n \geq 3$ and $\alpha > 0$. Suppose*

$$\Sigma(K) \geq \alpha n |K| \quad \textit{for every } K \in \mathcal{F}_n. \tag{6}$$

Then every inclusion-maximal Condorcet domain D on n alternatives satisfies

$$\Sigma(D) \geq \frac{(1+\alpha)n+1}{4}|D|.$$

Consequently,

$$f(n) \leq \frac{4n}{(1+\alpha)n+1} f(n-1), \qquad h(n) \leq \alpha^{-1} h(n-1).$$

Proof. A maximal domain is full. The component identity gives $\Sigma(D)+G(D) \geq \alpha n|D|$; Theorem 3.1, proved next, bounds the left side by $4\Sigma(D) - (n+1)|D|$. Apply the resulting inequality to a domain of size $f(n)$ and use $\Sigma(D) \leq n f(n-1)$. A maximum peak-pit domain can be completed under one chosen extreme law per triple. The completion is still peak-pit, so maximal size makes it equal to the original domain. It belongs to $\mathcal{F}_n$, and (6) gives the second recurrence. □

This is an algebraic formulation of the present argument. Any stronger uniform density estimate for this same extreme-law class would improve the unrestricted recurrence through the displayed formula. It does not require a change to the component reduction or gap inequality.

3 Moros's gap inequality

Theorem 3.1 (Gap inequality)**.** *For every inclusion-maximal Condorcet domain D on $n \geq 3$ alternatives,*

$$G(D) \leq 3\Sigma(D) - (n+1)|D|. \tag{7}$$

We use Moros's shorter proof [4, Sections 3–4]: two exact counts reduce the inequality to an injection from free alternatives into locked adjacent pairs. Let D be a set of orders and $\sigma = s_1 \cdots s_n \in D$. A *lift* of x in σ is an order of D obtained by moving x upwards and keeping the other alternatives in their relative order; a *drop* is defined symmetrically. The *lowest lift* is the lift that places x lowest, and the *highest drop* the drop that places it highest. We call x *blocked* in σ

if it has no lift, *free* if it has a lift and a drop, and *frozen* if it has neither. Letter $j \in [n-1]$ of σ is the adjacent pair (s_j, s_{j+1}). If $\mathrm{swap}_j(\sigma) \in D$, it is a lift of s_{j+1} and a drop of s_j. Letter j is *locked* if s_{j+1} is blocked and s_j has no drop: neither can pass the other. Let $\mathrm{fr}(D)$ and $\mathrm{fz}(D)$ be the numbers of pairs (σ, x) with x free, respectively frozen, in $\sigma \in D$, and $\mathrm{lk}(D)$ the number of locked letters (σ, j). Recall that $A(D)$ counts the pairs (σ, j) with $\mathrm{swap}_j(\sigma) \in D$.

Within one fibre of x, the member placing x highest is the only one in which x is blocked, and the member placing it lowest is the only one in which x has no drop. Summing over all fibres (cf. [1]),

$$\Sigma(D) = \#\{(\sigma, x) : x \text{ is blocked in } \sigma\} = \#\{(\sigma, x) : x \text{ has no drop in } \sigma\}. \tag{8}$$

Lemma 3.2. *Let D be a full domain, $\sigma \in D$ and $\mathrm{swap}_j(\sigma) \notin D$. Then s_{j+1} is blocked in σ or s_j has no drop in σ.*

Proof. Write $u = s_j$ and $w = s_{j+1}$. If fixed pairs are present, violating their comparison blocks both moves. Otherwise the swap changes only the comparison of u and w, so it violates the law of some triple $T = \{u, w, y\}$, and y is above both or below both. If y is below, σ and $\mathrm{swap}_j(\sigma)$ give uwy and wuy on T, so the law is wN1 or uN2. A lift of w also gives wuy, so w has none. If y is above, they give yuw and ywu, so the law is uN3 or wN2. A drop of u also gives ywu, so u has none. □

Lemma 3.3. *For every finite set D of orders,*

$$2\Sigma(D) = n|D| - \mathrm{fr}(D) + \mathrm{fz}(D),$$

and if D is a full domain, also

$$2\Sigma(D) = (n+1)|D| - A(D) + \mathrm{lk}(D).$$

Proof. By (8), $2\Sigma(D)$ counts the pairs (σ, x) in which x is blocked plus those in which x has no drop. A frozen pair is counted twice, a free pair never, and every other pair once; this is the first identity. For the second, give each blocked s_{j+1} and each s_j without a drop to letter j of σ. Only the top of each order, which is blocked, and the bottom, which has no drop, are left over. A letter with $\mathrm{swap}_j(\sigma) \in D$ receives nothing and a locked letter receives two. By Lemma 3.2 every other letter receives exactly one. So $2\Sigma(D) = 2|D| + (n-1)|D| - A(D) + \mathrm{lk}(D)$. □

The first identity is the sum over x of Fathi's fibre identity [1, eq. (96)]; the summed form appears in the proof of [1, Theorem K.189]. Adding the two identities and using Lemma 2.1, $A(D) = n|D| - \Sigma(D) - G(D)$, gives for every full domain

$$3\Sigma(D) - (n+1)|D| - G(D) = \mathrm{fz}(D) + \mathrm{lk}(D) - \mathrm{fr}(D). \tag{9}$$

So Theorem 3.1 follows once the free pairs are sent injectively into the locked letters. The second identity and Lemma 2.1 also give $\Sigma(D) = |D| + G(D) + \mathrm{lk}(D)$ for every full domain.

The injection uses maximality through the following observation.

Lemma 3.4. *Let D be a maximal Condorcet domain in which p is above q in every order. Then no order of D has p directly above q.*

Proof. Let $\rho \in D$ have p directly above q, and let ρ' be ρ with p and q exchanged; $\rho' \notin D$. On every triple $\{p, q, y\}$, D satisfies pN3 and qN1, and ρ' gives yqp or qpy, which satisfy qN1, respectively pN3. Other triples are ordered by ρ' as by ρ. So $D \cup \{\rho'\}$ is a Condorcet domain, against maximality. □

Lemma 3.5. *Let D be a Condorcet domain, $b = s_k$ free in $\sigma \in D$, $a = s_{k-1}$, $c = s_{k+1}$ and $T = \{a, b, c\}$. Then D satisfies aN3 or cN1 on T. If D satisfies aN3 on T, then in every lift of b the alternative a is directly above c, and the letter (a, c) is locked. If D satisfies cN1 on T, the same holds in every drop of b. If D is maximal, it does not satisfy both.*

Proof. On T, σ gives abc, a lift of b gives bac and a drop gives acb. So b takes every position, a only the first two and c only the last two; the only laws these orders allow are aN3 and cN1. In a lift τ of b, the alternatives from the new place of b down to a move down one place, so a is directly above c. A lift of c or a drop of a in τ would put c above a, with b still above a: a last on T. In a drop τ of b, a is directly above c and b is below c; a lift of c or a drop of a in τ would put c first on T. Finally, the orders of T allowed by both laws are abc, acb and bac, so a would be above c in every order of D, while a lift of b has a directly above c, against Lemma 3.4. □

The first assertion is [1, Lemma K.52]. Now let D be maximal. For a free pair (σ, b), with a, c, T as in Lemma 3.5, let τ be the lowest lift of b if D satisfies aN3 on T, and the highest drop of b otherwise. Put $\psi(\sigma, b) = (\tau, j)$, where j is the position of a in τ: the letter (a, c) of τ.

Proposition 3.6. *If D is a maximal Condorcet domain, ψ maps the free pairs of D injectively to its locked letters. So $\mathrm{fr}(D) \le \mathrm{lk}(D)$.*

Proof. The images are locked letters by Lemma 3.5. Let (τ, j) be a locked letter, $p = s_j(\tau)$, $q = s_{j+1}(\tau)$, and (σ, b) a preimage. Then $b \notin \{p, q\}$, and σ is τ with b moved into the slot between p and q; so b determines the preimage. If $\psi(\sigma, b)$ used a lift, then b is above p in τ, D satisfies pN3 and not qN1 on $\{p, q, b\}$, and b has a drop in σ; call b *upper*. Otherwise b is below q in τ, D satisfies qN1 and not pN3 on $\{p, q, b\}$, and b has a lift in σ; call b *lower*. For upper b, some $\chi_b \in D$ puts q first on $\{p, q, b\}$, and as p is never last there, χ_b gives qpb. Suppose that $b_1 \neq b_2$ give preimages (σ_1, b_1) and (σ_2, b_2).

Both upper. Let ρ_k be a drop of b_k in σ_k. On $\{b_1, b_2, q\}$, $\sigma_1, \sigma_2, \rho_1, \rho_2$ give b_2b_1q, b_1b_2q, b_2qb_1, b_1qb_2. So b_1 and b_2 take every position, and D satisfies qN1 there. In χ_{b_1}, q is above b_1, hence below b_2, so χ_{b_1} gives b_2qp on $\{p, q, b_2\}$, against pN3.

Both lower. Reversing every order keeps D a maximal Condorcet domain, exchanges lifts and drops (the lowest lift becomes the highest drop) and maps locked letters to locked letters. As D never satisfies both laws of Lemma 3.5, the rule defining ψ is symmetric, so reversal maps ψ to ψ and exchanges upper and lower. Apply the previous case.

b_1 upper, b_2 lower. On $\{p, q, b_1, b_2\}$, τ gives b_1pqb_2, σ_1 gives pb_1qb_2, σ_2 gives b_1pb_2q, and $\chi = \chi_{b_1}$, in which q is above p, has b_2 above q (by qN1 on $\{p, q, b_2\}$): it gives b_2qpb_1. Let ρ be a drop of b_1 in σ_1 and λ a lift of b_2 in σ_2. On $\{b_1, b_2, q\}$, τ, σ_2, χ give b_1qb_2, b_1b_2q, b_2qb_1, and ρ puts q first. So b_2 and q take every position, D satisfies b_1N2 there, ρ gives qb_2b_1 and λ gives b_1b_2q. On $\{p, b_1, b_2\}$, $\tau, \sigma_1, \rho, \lambda, \chi$ then give b_1pb_2, pb_1b_2, pb_2b_1, b_1b_2p, b_2pb_1: each of p, b_1, b_2 takes every position, which is impossible in a Condorcet domain. □

Proof of Theorem 3.1. A maximal domain is full, so (9) holds, and its right-hand side is nonnegative by Proposition 3.6. □

4 A finite window inequality

For $H \in \mathcal{F}_k$, its *bottom slices* group orders by their last alternative and delete it; top slices group by the first alternative. Every nonempty slice belongs to $\mathcal{F}_{k-1}$. Write $a_j(H)$ for the number of orders

whose swap at position j is enabled. On seven labels put

$$C(H) = \frac{a_1 + 2a_2 + 2a_3 + 2a_4 + 2a_5 + a_6}{10}. \tag{10}$$

The weights sum to one. They are chosen so that each interior swap is counted with total weight one when a seven-position window slides through an order.

Lemma 4.1 (Exact finite estimate)**.** *There is a rational function q on the isomorphism types of $\mathcal{F}_6$ such that, for every $J \in \mathcal{F}_7$,*

$$C(J) \le \frac{5}{8}|J| + \sum_{H \text{ bottom slice}} q(H) - \sum_{H \text{ top slice}} q(H). \tag{11}$$

The rational table and exact verifiers are supplied in the certificate supplement. The finite statement includes all fixed-pair constraints, with no common-axis assumption.

Proof. We separate the certificate, its exceptional receiver case and its coverage. The supplement gives the complete tables and replay instructions.

Certificate. The certificate assigns per-order weights ϕ to five-label types. For a six-domain H let $b(H)$ and $t(H)$ be the sums of $|L|\phi(L)$ over its bottom and top five-slices. Put

$$s(H) = \frac{12}{19}|H| + b(H) - t(H) - \frac{1}{5}A(H), \qquad \eta = \frac{1}{152}.$$

There are specified exceptional sets $\mathcal{T}, \mathcal{S}$ of five and nine six-label types. Put $p(H) = 2\eta(r_t(H) - r_b(H))$ outside $\mathcal{S}$, and $p(H) = \psi(H)$ on $\mathcal{S}$, with the rational values ψ supplied by the certificate. Set $\theta_b = s/2 + p$ and $\theta_t = s/2 - p$. The finite facts are as follows. On $\mathcal{T}$ both split values are zero. Outside $\mathcal{T} \cup \mathcal{S}$,

$$\theta_b(H) \ge \frac{\eta}{2}|H| + 4\eta r_t(H), \qquad \theta_t(H) \ge \frac{\eta}{2}|H| + 4\eta r_b(H). \tag{12}$$

Here r_b, r_t count specified five-slices described below. A further finite check proves

$$\Lambda(J) := \sum_{\text{bottom } H} \theta_b(H) + \sum_{\text{top } H} \theta_t(H) \ge \eta|J| \tag{13}$$

whenever J has a slice in $(\mathcal{T} \setminus \{R_8\}) \cup \mathcal{S}$. All types, weights and slice selectors are explicitly defined in the ancillary data; they are not fitted during verification.

The remaining exceptional type. The only type not handled by the explicit seven-label check is

$$R_8 = \{524130, 524310, 534120, 534210, 542130, 542310, 543120, 543210\}.$$

It has a common first and last alternative. The quantity $r_b(H)$ counts bottom five-slices of the type R_8 with its first alternative deleted; $r_t(H)$ counts top five-slices of the type with its last deleted. Both counts vanish on R_8. If a bottom R_8 slice occurs, its eight orders share their first alternative, so they enter one top receiver slice and contribute one to its r_b. That receiver is not R_8. Distinct bottom slices give distinct receiver/label pairs. Each contributes 4η, exactly paying the missing baseline $8\eta/2$. Top R_8 slices are symmetric. In the remaining cases all non-R_8 slices lie outside $\mathcal{T} \cup \mathcal{S}$, so (12) applies to every receiver. It proves (13) in every case not handled by the exceptional finite check.

Define

$$q(H) = \frac{b(H) + t(H)}{2} + p(H).$$

In a seven-domain, a top slice of a bottom slice is the same conditional five-domain as the corresponding bottom slice of a top slice. These middle terms cancel. Also the sum of $A(H)$ over all six-slices is $10C(J)$. Consequently

$$\Lambda(J) = \frac{12}{19}|J| - C(J) + \sum_{\text{bottom } H} q(H) - \sum_{\text{top } H} q(H).$$

Now (13) and $12/19 - 1/152 = 5/8$ give (11).

Coverage. We finally explain why the finite checks are exhaustive. All 128 400 labelled five-domains form 1 160 relabelling types. Extend their defining laws over the ten new triples and five new pairs. This gives 14 105 084 nonempty six-extensions of five-type representatives and 94 830 feature rows. The extension count is not a count of distinct labelled six-domains. Every six-domain occurs up to relabelling: restrict its *defining conditions* to five labels, rather than taking its projection. Keeping every domain on the thirteen exceptional feature rows gives precisely the fourteen required types. Two of the five types in $\mathcal{T}$ share one feature row. Their representatives and reversals, omitting R_8, give eighteen seven-label seed families. Requiring a prescribed endpoint six-slice and intersecting all compatible law and pair options produces 14 248 seed–domain cases (before deduplication across seeds) and 18 316 feature rows, covering both orientations. Every required inequality is verified by clearing the rational denominators and checking a nonnegative integer residual. A separate tuple-set checker recomputes every saved row from its witness orders. The coverage and receiver arguments above are mathematical parts of the proof; exact arithmetic verifies the finite residuals. □

5 Cancellation, concatenation and the bounds

Proposition 5.1. *Every $K \in \mathcal{F}_n$ satisfies $\Sigma(K) \geq 3n|K|/8$.*

Proof. For $n \geq 7$, condition on the entire order outside each block of seven consecutive positions. The conditional domain belongs to $\mathcal{F}_7$: internal triples retain their extreme laws and crossing triples can only impose pair comparisons or constant restrictions. Apply Lemma 4.1 to all these windows, with their outside multiplicities. After regrouping by the exterior, the bottom six-slices of a window starting at position i are the top six-slices of the window starting at $i-1$. Thus every interior potential cancels. Since $\mathcal{F}_6$ is finite, $|q(H)| \leq Q|H|$ for a constant Q; the two endpoint layers contribute at most $2Q|K|$.

The weights in (10) count each interior enabled swap once. At the ends they miss at most five incidences per order: the total coefficient deficit is $(n-1) - (n-6) = 5$. Therefore

$$A(K) \leq \frac{5}{8}(n-6)|K| + (5+2Q)|K| \leq \frac{5}{8}n|K| + B|K|$$

for a fixed B. By Lemma 2.1, $\Sigma(K) \geq (3n/8 - B)|K|$.

The boundary loss disappears by taking ordered copies. Fix any $K \in \mathcal{F}_n$, with M orders, and concatenate t copies on disjoint blocks, every alternative of an earlier block above every alternative of a later block. Fixed cross-block pairs and compatible extreme laws make this a member $K^{[t]} \in \mathcal{F}_{tn}$. Directly,

$$|K^{[t]}| = M^t, \qquad \Sigma(K^{[t]}) = t\Sigma(K)M^{t-1}.$$

Apply the affine bound to $K^{[t]}$ and divide by tM^{t-1}: $\Sigma(K) \geq (3n/8 - B/t)M$. Let t tend to infinity. This also covers the original small values of n. □

Proof of Theorem 1.1. Proposition 5.1 supplies (6) with $\alpha = 3/8$. Proposition 2.3 gives the two recurrences. Iterate from $f(3) = h(3) = 4$ to obtain the exponential bounds.

Ordered concatenation gives supermultiplicativity of both maxima. Fekete's lemma applied to their logarithms proves existence of the stated limits. Finally, retaining the factor $n/(n + 8/11)$ gives, for $n \geq 3$,

$$f(n) \leq 4(32/11)^{n-3} \prod_{k=4}^{n} \frac{k}{k + 8/11}.$$

The logarithm of the product is $-(8/11)\log n + O(1)$, which proves the polynomial correction (the product is empty when $n = 3$). □

Remark 5.2. The certificate was discovered with computational assistance; its verification uses exact rational arithmetic and no optimizer. The supplement distinguishes saved-data replay from complete regeneration. The Lean source, build instructions and precise verification scope are documented separately in the accompanying GitHub repository and its archived release [9].

Contributions and verification

Moros introduced the deletion-and-gap framework, the earlier explicit general bases, the peak-pit base 8/3 and the window/potential method [4]. Building on this, Riis developed the improved unrestricted bound and further mathematical refinements presented here, notably the 32/11 bound. The work was developed interactively with AI assistance from Claude and ChatGPT/Codex. The scope of Lean verification is documented in the accompanying repository [9].

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